

Auto-Generated Personas: Enhancing User-centered Design Practices among University Students

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ABSTRACT

Personas are commonly used in User-centered Design (UCD) activities to help designers better understand users' needs. However, there is still a reliance on traditional approaches such as interviews and ethnography for building personas in UCD activities. To this end, we developed an auto-generating persona system to enhance practices in UCD course activities. Our persona system is developed based on the GPT-4 model, the DALL-E 2 model, and knowledge graphs. Hence, our persona system includes three main features of our persona system: automated processing of survey data, automatic generation of 2D avatars, and providing options for automatic or customized entity generation. We recruited a total of 22 participants to evaluate our persona system. Our findings confirmed that there was a significant difference in terms of efficiency, satisfaction, accuracy, and diversity. Meanwhile, participants provided both positive and negative feedback regarding our persona system. As on-going work, we discuss the current limitations of our persona system and explore future research directions to further improve its capabilities and effectiveness.

CCS CONCEPTS

• **Human-centered computing** → *Interaction design*; **User centered design**; *Interaction design*; **Systems and tools for interaction design**;

KEYWORDS

Personas, Knowledge Graphs, GPT-4, User Studies

ACM Reference Format:

Xishuo Zhang, Lin Liu, Yi Wang, Xiao Liu, Hailong Wang, Chetan Arora, Haichao Liu, Weijia Wang, and Thuong Hoang. 2024. Auto-Generated Personas: Enhancing User-centered Design Practices among University Students. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '24)*, May 11–16, 2024, Honolulu, HI, USA. ACM, New York, NY, USA, 7 pages. <https://doi.org/10.1145/3613905.3651043>

1 INTRODUCTION

Personas are a popular tool created and used in User-centered Design (UCD) activities to better understand and represent users' needs in design processes [20]. It represents the different user types that are designed based on the analysis of user research data [27]. In UCD activities, personas emphasize the relationship between users and designers, ensuring that the products designed by the designers meet the needs of the target user group throughout the whole design iteration process [13]. In (human-centric) requirement engineering (RE) activities, personas serve as a key technique for better understanding demographic information, user needs, behaviors, motivations, and pain points [19, 22, 28]. Requirements engineers can design software systems based on personas to meet the needs of target users and ensure that the software system is user-friendly. Furthermore, personas are an important component

of knowledge in undergraduate courses, such as UCD and RE. Students frequently use personas in discussions and collaboration for user interface (UI), user experience (UX) and system design [2]. However, the accuracy of personas has been a paradox [9]. This issue is partly due to the lack of professional analysis processes by students, limited access to real users, and insufficient industry experience, which can lead to the creation of personas with low credibility [1, 2, 9]. Meanwhile, some practitioners (e.g., software developers) may lack the professional knowledge to build personas, a challenge that is particularly pronounced for practitioners from startups [30].

In recent years, there have been a growing number of new approaches and technologies for building personas, such as data-driven persona based on media data [17], the persona system based on mouse-tracking [18], the persona design guidelines tool [22], and persona based on knowledge graphs [32]. However, despite their potential, there is a lack of demonstrated value and effectiveness in integrating persona systems into UCD course activities. Additionally, with the rapid development of Large Language Models (LLMs) such as ChatGPT, the use of LLMs to develop personas is an important application in several fields, e.g., software engineering, UI design and education [3, 5, 24]. For instance, students can use tools like ChatGPT to assist in creating personas [12]. However, there is currently a lack of a system that automatically generates personas based on LLMs.

To this end, we developed an automated persona generation system to enhance students' practices in UCD course activities. There are three main features of our persona system: (1). Our persona system can automatically process survey data. (2). Our persona system can automatically generate persona templates and (3). Our persona system customizes or automatically creates entities. To evaluate our persona system, we deployed it in an undergraduate UCD course at the School of Design. We recruited a total of 22 participants, all of whom have backgrounds in design and art. We designed two sets of questionnaires: the preliminary persona questionnaire and the post-persona system questionnaire. The preliminary persona questionnaire aimed to understand persona creation using traditional approaches, which involve creating personas based on surveys and ethnographic approaches in the UCD course. The post-persona questionnaire was to evaluate persona creation using our persona system in the UCD course. The questionnaire covers six aspects, including efficiency, satisfaction, collaboration, accuracy, creativity, and diversity. Our results found that there was a significant difference in efficiency, satisfaction, accuracy, creativity, and diversity. Furthermore, we conducted a semi-structured interview with each participant. Our findings confirmed that many participants provided positive feedback about using our persona system, such as its efficiency and accuracy. Conversely, they also provided negative feedback, such as unfriendly UI/UX. Finally, we discussed the findings, current limitations, and future work.

2 RELATED WORK

2.1 Personas in Education

Personas are frequently used in the field of education, particularly in design and software engineering (SE) education. Cooper [8] was

the first to introduce the use of personas in SE, especially in requirements engineering (RE) activities, as a means to better understand the needs of end-users and other human aspects [19]. Palacin-Silva et al. [26] reported that personas can help in understanding user needs and preferences, and providing a foundation and inspiration for exploring application ideas in a SE capstone course. Arora et al. [4] reported a persona-based assessment of SE student research projects, finding that the persona-based assessment addressed diversity issues in research projects, student skills, and supervisor quality in SE research units. Furthermore, personas are an important tool and knowledge component in UCD. Anvari et al. [2] considered user-centered solutions as crucial in large classes. Students use personas to conceptualize design solutions and find that the quality of the final design artifacts is higher than those not using personas. They also found that personas can help in understanding the importance of cultural differences. Garreta-Domingo et al. [11] found that the use of persona-based teaching improves student collaboration in human-centered design activities, but also identified challenges and misconceptions faced by students in creating personas, such as lack of knowledge and experience, leading to improper use.

2.2 Existing Persona Systems and Approaches

Data-driven personas, interviews, and ethnography are the main methods for building personas [9]. Data-driven personas require extensive user data (e.g., behavioral, and demographic data), sourced from social media or other online platforms [23]. The interviews or ethnography are the commonly used approaches in UCD [7]. Chapman and Milham [7] found that interviews and ethnography cannot validate the effectiveness of data-driven personas. In the context of heuristic evaluation, personas in heuristic evaluation can help in understanding user needs, preferences and keeping evaluators user centered [10]. Heuristic evaluation can be traced back to 1990, when Nielsen and Molich first published their foundational heuristics [25]. In recent years, there has been an increased focus on developing persona systems. Jung et al. [18] proposed an interactive persona system based on mouse-tracking, which can precisely track user behaviors and interact with personas. It can also help with remote user research. Jung et al. [17] also proposed a novel method that can automatically generate personas from large-scale, real-time, and aggregate user data via online social media. meanwhile, with the rise of Large Language Models (LLMs), more researchers are focusing on using LLMs to create personas [16]. For example, Wang et al. [31] proposed a novel approach that uses LLMs as cognitive collaborators by dynamically identifying persona roles and engaging in multi-round self-collaboration. Zhang et al. [32] proposed a tool that can generate persona based on LLMs and knowledge graphs. However, this work still has many limitations and does not provide an evaluation for this tool. Therefore, our goal is to develop a persona system that can automatically generate personas and explore its potential value in enhancing UCD course activities.

3 THE OVERVIEW OF PERSONA SYSTEM

Our persona system is a web application to facilitate design major students in creating personas creation during their UCD course

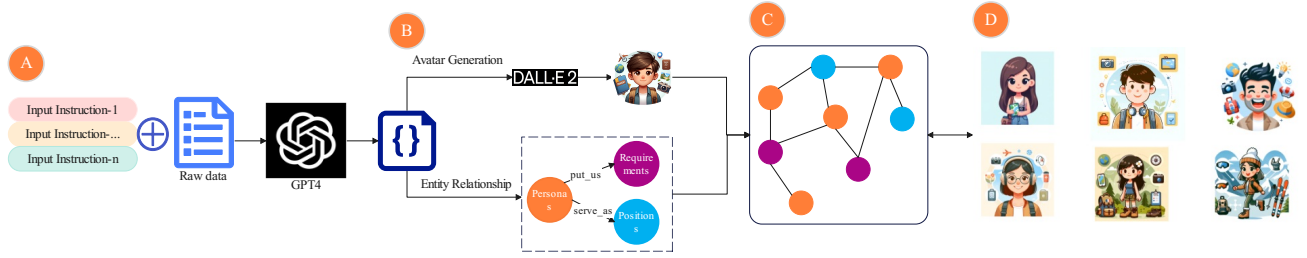


Figure 1: The overview of the persona system. (A) survey data processing, (B) automatic generation of 2D avatars & customization of knowledge graph nodes, (C) an example of knowledge graph nodes, and (D) generated persona templates.

activities. In this section, we describe the main features and implementation of our persona system. Figure 1 shows our persona system overview.

Our persona system consists of three main features: (1). Our persona system uses the GPT-4 model to automatically process the survey data. (2). Our persona system can automatically generate 2D avatars based on the processed survey data. Our persona system also allows for the customization of additional commands for 2D avatar generation. (3). Our persona system can automatically generate entities or allow for the customization of entities, and it displays the knowledge graph visualization within the persona system.

3.1 System Implementations

As for this implementation, we developed the front-end of our persona system using HTML¹, CSS², and a JavaScript-based open-source framework Vue.js³. For the back-end development, we used Java to interface with APIs of the GPT-4⁴, which serves as the core module for processing survey data and customizing commands. We also developed the functionality to upload and parse survey data and transfer the parsed data to the GPT-4 model.

For the automatic 2D avatar generation, we developed a feature that can automatically generate 2D avatars based on demographic information using DALL-E 2⁵. Our persona system also supports the addition of custom commands to generate preferred 2D avatars.

For the knowledge graphs, we developed an entity extraction model using Python and BERT⁶, with each entity representing a module of the persona. For example, Figure 2 shows the section A - E representing different types of entities. Our persona system supports the customization of entities, and it can also automatically create entities based on processed survey data. We used the neo4j⁷ database to manage and store the extracted entity data. Figure 3 shows the knowledge graph visualization through neo4j.

4 USER STUDY

We conducted a preliminary evaluation of the persona system with regards to user experience and feedback. The Research Ethics Committee at our university reviewed and approved our study.

4.1 Participants

We recruited 22 participants (7 males and 15 females) from the local campus with ages ranging from 20 to 23 ($Mean = 21.23$, $SD = 0.90$). All participants were enrolled in the UCD course at the university (school of design) with prior background in design and persona creation.

4.2 Procedure

All participants were enrolled in UCD from September 2023 to December 2023. During UCD course activities, the research introduced the key methods of UCD for collecting user data, such as focus groups, interviews, and questionnaires [6]. The researcher introduced the methods for creating personas, which included traditional approaches and our persona system. The terms “*traditional approaches*” refer to survey and ethnographic approaches. After the two-month-long UCD course, all participants collected survey data using online survey platforms like Wenjuanxing⁸ and Tencent Questionnaire⁹ and analyzed the data using their preferred approaches to build personas. Additionally, all participants could also use LLMs such as ChatGPT [12], for data analysis and persona building. After completion of their persona creation, all participants were required to fill out a preliminary persona questionnaire.

The researcher then introduced the usage of the processes of the persona system. Next, all participants used the collected survey data to generate persona templates through our persona system, which served as analytical and collaborative tools to discuss users’ motivations, needs, human factors, etc. After participants completed the discussion and activities, they were required to fill out a post-persona system questionnaire. All participants also voluntarily took part in a semi-structured interview lasting for about 15 minutes which was audio-recorded. The study lasted for about 60 minutes in total. We provided an e-gift card for each participant as an incentive in local currency.

¹<https://www.runoob.com/html/html5-intro.html>

²<https://www.runoob.com/css3/css3-tutorial.html>

³<https://cn.vuejs.org/>

⁴<https://openai.com/gpt-4>

⁵<https://openai.com/dall-e-2>

⁶https://huggingface.co/docs/transformers/model_doc/bert

⁷<https://neo4j.com/>

⁸<https://www.wjx.cn/>

⁹<https://wj.qq.com/>

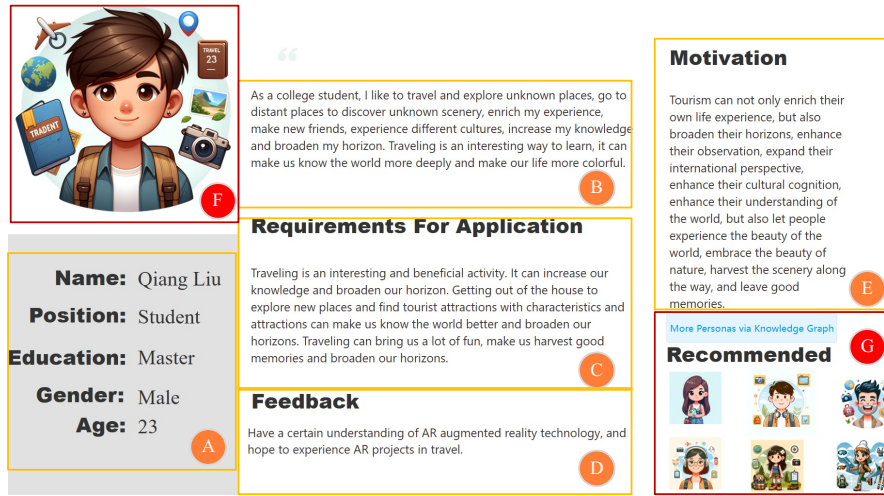


Figure 2: An example of generating persona template via persona system. (A) - (E) represents the customization of knowledge graph nodes, (F) automatic generation of a 2D avatar-based processed survey data, and (G) related personas recommendation through knowledge graphs.

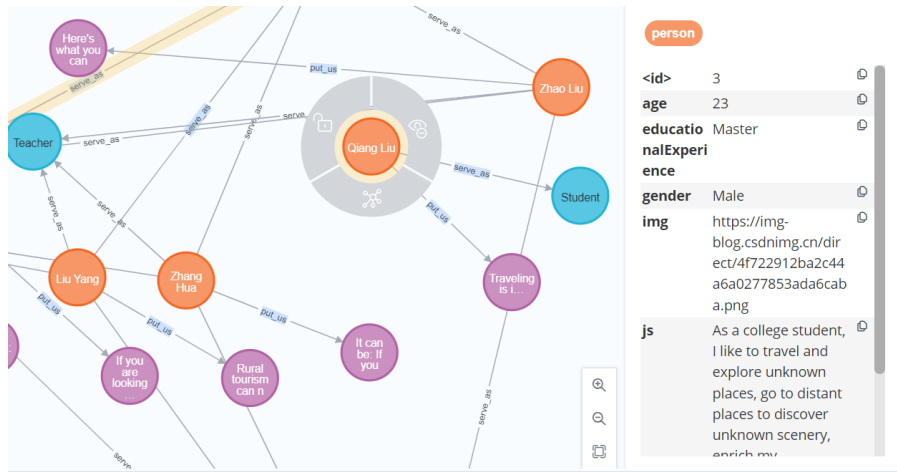


Figure 3: Visualization of the knowledge graphs.

4.3 Measures

4.3.1 Preliminary Persona Questionnaire. Participants were required to fill out a preliminary persona questionnaire following their practices in UCD course activities. The preliminary persona questionnaire includes six questions, each rated on a 7-point Likert scale (1: Strongly Disagree to 7: Strongly Agree). The preliminary persona questionnaire asked participants to answer about efficiency, satisfaction, collaboration, accuracy, creativity, and diversity. Furthermore, the preliminary persona questionnaire contained questions about participants' demographic information.

4.3.2 Post-Persona System Questionnaire. After using our persona system, participants were asked to submit a post-persona questionnaire with the same questions are preliminary questionnaire. All the questionnaires were in Chinese.

4.3.3 Semi-structured Interview. After completing the activities and experiment, we conducted a semi-structured interview asking about overall experiences and feedback about our persona system. The semi-structured interview contained seven main questions. The questions asked participants to answer the challenges of creating personas, satisfaction with our persona system (including design, architecture, and knowledge graph visualization), the accuracy of the persona they created, and the current limitations of using our persona system. The interview was conducted in Chinese.

5 FINDINGS

We applied Wilcoxon Signed-Ranked tests for the preliminary persona questionnaire and the post-persona system questionnaire. We

applied affinity diagramming for organizing the participants' feedback [14]. The first author and two co-authors also translated the transcript of the interview into English and checked data accuracy.

5.1 Preliminary Persona Questionnaire vs. Post-Persona System Questionnaire

Participants were asked about their experiences using traditional approaches and our persona system to create personas during their practices in the UCD course activities. Figure 4 shows the rating results. Below, we show results from each question in the preliminary persona questionnaire and post-persona system questionnaires.

Efficiency. In terms of efficiency, participants rated the use of the persona system more highly ($M = 5.00$, $SD = 1.41$) than the traditional approaches ($M = 4.00$, $SD = 1.27$). Meanwhile, there was a **significant difference** between using the persona system and the traditional approaches ($p = .04$).

Satisfaction. In terms of satisfaction, participants rated the use of the persona system more highly ($M = 4.81$, $SD = 1.68$) than the traditional approaches ($M = 3.68$, $SD = 1.32$). Meanwhile, there was a **significant difference** between using the persona system and the traditional approaches ($p = .03$).

Collaboration. In terms of collaboration, participants gave a similar rating for both the use of the persona system ($M = 5.09$, $SD = 1.50$) and the traditional approaches ($M = 5.00$, $SD = 1.44$). Furthermore, there was no significant difference between using the persona system and the traditional approaches ($p = .80$).

Accuracy. In terms of accuracy, participants rated the use of the persona system more highly ($M = 5.18$, $SD = 1.29$) than the traditional approaches ($M = 3.81$, $SD = 1.46$). Meanwhile, there was a **significant difference** between using the persona system and the traditional approaches ($p < .01$).

Creativity. In terms of creativity, participants rated the use of the persona system more highly ($M = 5.09$, $SD = 1.50$) than the traditional approaches ($M = 4.22$, $SD = 1.37$). However, there was no significant difference between using the persona system and the traditional approaches ($p = .06$).

Diversity. In terms of diversity, participants rated the use of the persona system more highly ($M = 4.77$, $SD = 1.90$) than the traditional approaches ($M = 3.59$, $SD = 1.36$). Meanwhile, there was a **significant difference** between using the persona system and the traditional approaches ($p = .02$).

5.2 Qualitative Findings

We also summarized qualitative interview data to understand the positive and negative feedback from participants about our persona system.

5.2.1 Positive Findings. Some participants agreed with the contributions of our persona system ($N = 13$). Some participants mentioned that automatic persona generation could improve the efficiency of the work in user research ($N = 5$). Some participants considered that automatically generating personas was more accurate than using traditional approaches ($N = 18$). Some participants explained that there was no specialized data analysis process and most students likely relied on basic statistical results from survey systems and 'subjective analysis' as their method of analyzing

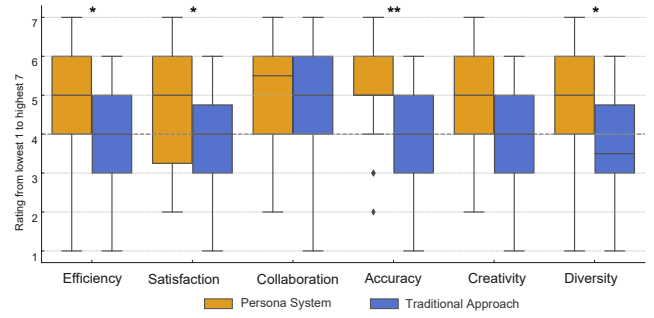


Figure 4: The figure shows the results of the preliminary persona questionnaire (represents the traditional approach) and post-persona system questionnaire (represents the persona system). ‘*’, ‘’, ‘***’ indicate $p < .05$ and $p < .01$ respectively. The grey line represents the median ($M = 4$).**

qualitative data ($N = 6$). Some participants also mentioned they overlooked the accuracy of the personas ($N = 9$). To ensure the integrity of UCD course activities, personas primarily serve a ‘demonstration’ role in this context.

We asked the participants about the visualization of relationship-based knowledge graphs. Some participants found the visualization of relationship-based knowledge graphs to be very useful ($N = 11$). Some participants considered that visualization of relationship-based knowledge graphs provided rich information about the relationship between different types of information, improving the understanding and analysis of relationships between personas and user needs during activities in the UCD course ($N = 5$). Furthermore, some participants considered that the persona system provided diverse information (e.g., software requirements) within the persona templates ($N = 9$).

5.2.2 Negative Findings. Some participants mentioned that the UI and UX could be improved in the future ($N = 20$). Specifically, participants considered that the persona system was inconvenient to operate and some steps were difficult to understand ($N = 9$). Some participants also mentioned that the system lacked a unique design style and that persona templates had a monotonous design style ($N = 11$). Some participants considered that the operation of the persona system was complex. They suggested optimizing some functions (e.g., data processes) and providing detailed tutorials ($N = 8$). Additionally, some participants mentioned a lack of understanding of knowledge graphs and the difficulty in understanding the relationships between difference nodes in the visualization of those relationship-based knowledge graphs ($N = 11$).

6 DISCUSSION

Here, we summarize the findings of the user study, discuss the positive and negative findings of the persona system, outline the current limitations, and recommend future work.

6.1 Improving Collaboration and Quality

Based on our quantitative findings, we found that there was no significant difference between the persona system and traditional

approaches in terms of collaboration. Thus, our persona system did not demonstrate a significant improvement in facilitating collaboration. Participants also found that the generated personas lacked creativity, which may be due to the system's limited support for diverse design styles and diversity and a variety of persona templates. This issue of limited design style and diversity was also highlighted in our qualitative findings by some participants. Additionally, a few participants gave a low rating about the accuracy of the auto-generated personas. They might find that the personas do not align with their perspectives. Some studies also exemplify that discrepancies between personas and developers' viewpoints can contribute to paradoxes in design and development processes [9]. Overall, we recommended that the persona system should be further developed to enhance collaboration among students, as well as between students and teachers, and help reduce the paradoxes arising from discrepancies between personas and students. Meanwhile, the persona system should provide a variety of specially designed, and high-quality persona templates to meet special user needs, such as accessibility personas [15].

6.2 Visualization with Personas

From our qualitative findings, we found that participants had a positive response to the visualization of relationships using knowledge graphs. Although we recruited participants from the School of Design, most participants were unfamiliar with the knowledge graphs before our study. As a result, some participants found the visualization of relationships based on knowledge graphs to be difficult to understand.

6.3 Personas in Education Activities

From our qualitative findings, we concluded that participants often focused on the design style of the persona templates, which sometimes led them to overlook important activities such as collaboration, brainstorming, and user need analysis based on personas [19, 28]. Additionally, we found that some participants lacked expertise in professional qualitative and quantitative data analysis. Although participants created personas using their preferred approaches, these personas might not effectively provide values in UCD course activities.

6.4 Limitation and Future Work

Our persona system has several limitations to be improved in future work. First, we plan to develop more visualization panels, including demographic information and requirement visualization [29, 32]. We also plan to enhance the visualization recommendations and classification accuracy of the knowledge graphs [21]. Meanwhile, we plan to explore the relationships and applications between visualizations, knowledge graphs, and personas in future educational activities. Second, different types of data may emerge in the iterations of designing software applications. We plan to develop comprehensive data processing capabilities, such as field research and interview data. Third, we plan to apply our persona system in requirements engineering education for software engineering (SE) students. Currently, how to apply personas in RE or human-centric requirements engineering (HCRE) is a significant research area in SE. Furthermore, participants mentioned that the persona

system was difficult to operate. Therefore, we plan to improve the design, experience, and functionality of our persona system, with a particular focus on interactions with the knowledge graph visualization, and evaluate its usability in the future. Finally, due to participants choosing their own methodologies, our results currently lack a standard of comparison across the study. Therefore, we plan to develop a more comprehensive evaluation method to examine the interactions between various methodologies and our persona system within the context of UCD course activities.

7 CONCLUSION

In this paper, we designed a persona system to enhance students' activities in the UCD course. Here, we designed two sets of questionnaires to evaluate six aspects (efficiency, satisfaction, collaboration, accuracy, creativity, and diversity) of both the persona system and traditional approaches in the UCD course activities. We also conducted a semi-structured interview with all participants to understand their positive and negative feedback. We recruited a total of 22 participants. Our findings confirmed the values and limitations of our persona system in UCD course activities. In future work, we plan to enhance the functionality and design of our persona system and explore its application value in the field of human-centric requirements engineering for SE students.

ACKNOWLEDGMENTS

This work is supported by the Key Laboratory of Infinite-dimensional Hamiltonian System and Its Algorithm Application (Inner Mongolia Normal University), Ministry of Education (Grand No.2023KFZD03, 2023KFYB03), the Natural Science Foundation of Inner Mongolia (2023LHMS06006), the Fundamental Research Funds for the Inner Mongolia Normal University (2022JBYJ032), and Archival Science and Technology Project of the Archives of Inner Mongolia Autonomous Region (2023-13) .

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